

Module Example Assignment

Chapter X, Sections Y-Z

Your Name

Date Assignment is Due

Problem Set

X.Y.1 Some exercise from the textbook that likely has inline latex like $\int_0^\infty e^{-x} dx$.

The answer to the exercise generally has inline latex as well like $\int_0^\infty e^{-x} dx$. Often times there are lines like the following which are all LaTeX,

$$\begin{aligned}\int_0^\infty e^{-x} dx &= [-e^{-x}]_0^\infty \\ &= \lim_{x \rightarrow \infty} (-e^{-x}) - (-e^0) \\ &= 0 - (-1) = 1.\end{aligned}$$

Software Assignment

Some prompt assigned as a software assignment. For example, create a visualization representing $\int_0^\infty e^{-x} dx$.

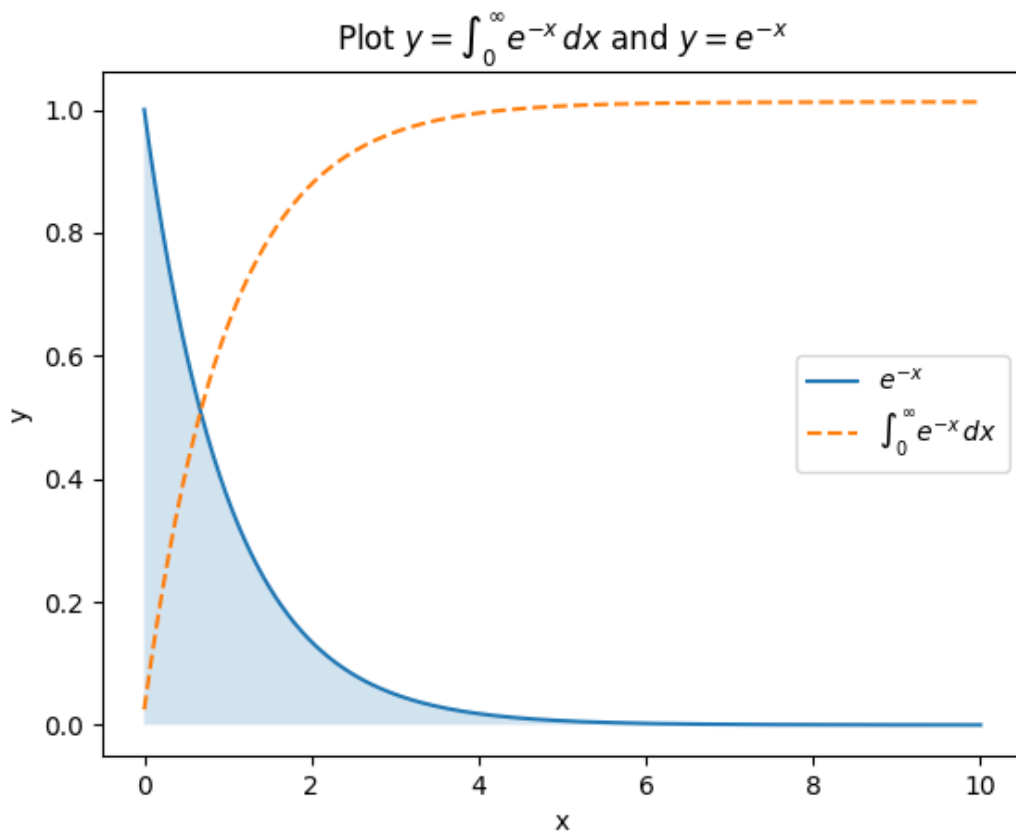


Figure 1: Visualization of e^{-x} and its integral approximation.

Below is the Python code used to generate the figure:

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Define the function  $e^{-x}$ 
5 def f(x):
6     return np.exp(-x)
7
8 # Generate x values from 0 to 10 (as infinity is not practical)
9 x = np.linspace(0, 10, 400)
10 y = f(x)
11
12 # Calculate the integral using numpy's cumulative trapezoidal rule
13 integral = np.cumsum(y) * (x[1] - x[0])
14
15 # Plot the function
16 plt.plot(x, y, label='$e^{-x}$')
17 plt.fill_between(x, y, alpha=0.2)
18
19 # Plot the integral
20 plt.plot(x, integral, label='$\int_0^{\infty} e^{-x} \, dx$', linestyle='--')
21
22 # Add labels and legend
23 plt.xlabel('x')
24 plt.ylabel('y')
25 plt.title('Plot  $y = \int_0^{\infty} e^{-x} \, dx$  and  $y = e^{-x}$ ')
26 plt.legend()
27
28 # Save the plot in the new directory
29 plt.savefig("/workspace/module-example-tex/software-assignment.png")
```